

CISS358: Algorithm Analysis
Assignment 3

OBJECTIVES

- Review of proof using induction and WOP.
- Prove the correctness of an algorithm using loop invariants.

Q1. Recall from class, I mentioned that there are some proofs that you must know because these proofs almost form a template for many other proofs. The goal of this question is not to design a proof, but rather to re-study a given proof very carefully and write an absolutely correct proof in your own words and possible filling gaps if necessary.

Prove the following using WOP.

EUCLIDEAN PROPERTY OF $\mathbb{N}$. If $a, b \in \mathbb{N} = \{0, 1, 2, \dots\}$ with $b > 0$, then there exist q and r in $\mathbb{N}$ such that

$$a = bq + r, \quad 0 \leq r < b$$

(More is true: q, r are unique. The proof is not difficult and is probably proven in discrete 1. I definitely talk about it in 451.)

SOLUTION.

$$r = a - bq \in X = \{a - bx \mid x \in \mathbb{Z}, a - bx \geq 0 \subseteq \mathbb{N}\}$$

$$\text{let } X = \{a - bx \mid x \in \mathbb{Z}, a - bx \geq 0\}$$

We will show X is a non empty set CASE I. $x = 0, a - b \cdot 0 = a - 0 = a \geq 0$ then $a \in X$ and X is non empty

By WOP X has a least element i.e $a - bx = r$

for some integer x say $x = q$

$$a - bq = r \rightarrow a = bq + r$$

by contradiction assume $r \geq b$ so, $0 \leq r - b$

$$0 \leq a - bq - b = a - b(q + 1)$$

so $a - b(q + 1) \in X$ i.e $r - b \in X$

but r is the least element in X

We will now prove the uniqueness of r, q

Assume $\exists r_1, q_1 \in \mathbb{Z} 0 \leq r_1 < b$ such that $a = a_1b + r_1$

$$\text{then } qb + r = q_1b + r_1$$

$$b(q - q_1) = r_1 - r$$

so $-b < b(q - q_1) < b$

$$-1 < q - q_1 < 1$$

so $q - q_1 = 0$ i.e $q = q_1$ hence $r = r_1$

□

Q2. Prove using invariants and induction that the following algorithm (bottom-up recursion for exponentiation) is correct.

```

ALGORITHM: power
INPUTS: x - base
        e - exponent
OUTPUT: x^e, i.e., x raised to power of e

ret = 1
while e > 0:
    if e % 2 == 1:
        ret = ret * x
    e = e / 2
    x = x * x

return ret

```

(Note: / is integer division.)

SOLUTION.

Let $P(e)$ be

$$P(e) = (ret = x^e, \text{ for some } x)$$

BASE CASE Let $P(0)$ be the invariant

Before the loop start $ret = 1$ and $x^0 = 1$ Hence $P(0)$ holds

INDUCTIVE CASE Assume $P(i)$ holds which means

$$P(i) = (ret = x^i, \text{ for some } x)$$

Consider $P(i+1)$

CASE I if $(i+1) \% 2 == 1, ret = x$

$$\lfloor \frac{i+1}{2} \rfloor = i/2, x = x * x$$

from $P(i)$ we know $P(\frac{i}{2})$ holds because $\frac{i}{2} < i$ for some x^2

$$\text{So } ret = x * (x^2)^{\frac{i}{2}} = x * x^i = x^{i+1}$$

CASE II if $(i+1) \% 2 == 0, ret = 1$

$$\lfloor \frac{i+1}{2} \rfloor = \frac{i+1}{2}, x = x * x$$

from $P(i)$ we know $P(\frac{i+1}{2})$ holds because $\frac{i+1}{2} < i$ for some x^2

$$ret = 1 * (x^2)^{\frac{i+1}{2}} = 1 * x^{i+1} = x^{i+1}$$

Hence by Weak Induction $P(e)$ holds

□

Q3. Prove that the selection sort algorithm is correct using invariants and induction.

SOLUTION.

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ALGORITHM: SELECTION_PASS
INPUTS: x - array,

minvalue = x[0]
minindex = 0

for i = 1, 2, ..., n-1:
    if x[i] < minvalue:
        minvalue = x[i]
        minindex = i
if minindex != 0:
    swap (x[0], x[minindex])
  
```

- a. Show that $x[0]$ is the minimum value of $x[0 \cdots n-1]$
- b. Show that $x[0 \cdots n]$ are the same as the values of $x[0 \cdots n-1]$ before the loop
- c. $x[x]$ is unchanged if k is not in $x[0 \cdots n-1]$

(a.) BASE CASE $P(0)$ the minimum of $x[0 \cdots 0]$

INDUCTIVE CASE Let $0 \leq i \leq n-1$ suppose $P(i)$ holds

Consider the $i+1$ st iteration

CASE I $x[i] < \text{minvalue}$ there is a swap

so $x[i]$ is the minimum value

CASE II $x[i] \geq \text{minvalue}$ there is no swap, so $x[\text{minvalue}]$

is the minvalue and there is a swap between the minvalue and $x[0]$ so $x[0]$

holds the minvalue

Hence $P(i+1)$ holds

(b.) for $-1 \leq i < n$ Let $Q(i)$ be the invariant

$Q(i) = (x[0 \cdots i-1])$ is a permutation of $x[0 \cdots i-1]$ at the begining

of the algorithm and $x[k]$ is unchanged for $k \notin x[0 \cdots i - 1]$)

BASE CASE Since only one value and nothing is done on the loop on the array $x[0 \cdots 0]$

Hence $Q(0)$ holds

INDUCTIVE CASE Suppose $Q(i)$ at $(i + 1)$ st operation

□